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End Semester Examination 2024-25

Name of the Program: B.Tech Biotechnology

Semester: III

Name of the Course: Biochemistry

Paper Code: **TBT- 304**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO 1
(a)	Draw the structure of polar uncharged amino acids and differentiate between essential and non-essential amino acids with examples.	
(b)	Describe the primary, secondary, tertiary and quaternary structure of proteins.	
(c)	Illustrate urea cycle with a diagram and the reactions that feed amino groups into the cycle.	
Q2	(20 marks)	CO 2
(a)	What are the inhibitors and uncouplers of oxidative phosphorylation?	
(b)	Discuss the fate of pyruvate under anaerobic conditions along with the reactions.	
(c)	Describe the various steps of the glycolysis pathway.	
Q3	(20 marks)	CO 3
(a)	Write a note on alpha and omega oxidation of fatty acids.	
(b)	Describe the oxidation of unsaturated and odd carbon fatty acids.	
(c)	Explain saponification value, iodine number, acid value and rancidity of fats.	
Q4	(20 marks)	CO 4
(a)	Write a note on how metals are involved in the catalysis of various enzymes.	
(b)	Explain the role of Vitamin B12 and NAD ⁺ /NADP ⁺ in enzyme catalysis.	
(c)	Differentiate between apoenzyme and holoenzyme with examples and explain why metals are good cofactors?	
Q5	(20 marks)	CO 5
(a)	Illustrate the structures of pyrimidines and purine nucleotides with a diagram.	
(b)	Discuss the diffraction and spectroscopic biophysics techniques.	
(c)	Explain the denovo biosynthesis of purine and pyrimidine nucleotides.	

